



VP AIR TEST NEET (2025-26)

PART TEST - 10

DROPPER

DURATION: 180 Minutes

Date: 01-03-2026

M. MARKS: 720

ANSWER KEY

1. (3)	46. (1)	91. (4)	136. (3)
2. (2)	47. (2)	92. (1)	137. (1)
3. (3)	48. (4)	93. (3)	138. (4)
4. (2)	49. (3)	94. (2)	139. (4)
5. (4)	50. (1)	95. (4)	140. (2)
6. (2)	51. (3)	96. (2)	141. (3)
7. (4)	52. (1)	97. (3)	142. (4)
8. (3)	53. (2)	98. (1)	143. (4)
9. (1)	54. (3)	99. (3)	144. (3)
10. (2)	55. (3)	100. (2)	145. (3)
11. (2)	56. (4)	101. (3)	146. (3)
12. (3)	57. (1)	102. (4)	147. (3)
13. (4)	58. (3)	103. (1)	148. (3)
14. (4)	59. (1)	104. (2)	149. (1)
15. (2)	60. (2)	105. (3)	150. (4)
16. (4)	61. (2)	106. (2)	151. (2)
17. (2)	62. (3)	107. (2)	152. (1)
18. (1)	63. (3)	108. (4)	153. (4)
19. (2)	64. (1)	109. (2)	154. (4)
20. (3)	65. (2)	110. (1)	155. (4)
21. (4)	66. (3)	111. (2)	156. (4)
22. (4)	67. (3)	112. (4)	157. (2)
23. (2)	68. (2)	113. (1)	158. (3)
24. (4)	69. (1)	114. (2)	159. (1)
25. (4)	70. (3)	115. (3)	160. (3)
26. (4)	71. (2)	116. (4)	161. (2)
27. (3)	72. (3)	117. (1)	162. (2)
28. (2)	73. (2)	118. (2)	163. (1)
29. (3)	74. (3)	119. (3)	164. (2)
30. (2)	75. (3)	120. (3)	165. (3)
31. (2)	76. (3)	121. (4)	166. (3)
32. (4)	77. (1)	122. (3)	167. (2)
33. (3)	78. (1)	123. (2)	168. (1)
34. (2)	79. (3)	124. (4)	169. (2)
35. (2)	80. (3)	125. (3)	170. (3)
36. (2)	81. (1)	126. (1)	171. (4)
37. (3)	82. (3)	127. (4)	172. (3)
38. (4)	83. (2)	128. (2)	173. (2)
39. (2)	84. (2)	129. (3)	174. (2)
40. (3)	85. (3)	130. (1)	175. (4)
41. (2)	86. (1)	131. (3)	176. (1)
42. (2)	87. (2)	132. (1)	177. (2)
43. (4)	88. (3)	133. (4)	178. (2)
44. (3)	89. (2)	134. (2)	179. (4)
45. (2)	90. (2)	135. (3)	180. (3)

HINTS & SOLUTIONS

1. (3)
According to Brewster's law:
 $\tan i_p = \mu = \sqrt{3} \Rightarrow i_p = 60^\circ$. Since $r + i_p = 90^\circ$,
the angle of refraction $r = 90^\circ - 60^\circ = 30^\circ$.
(NCERT Ed. 2025-26, 12th, Page No. 158)
2. (2)
Conservation of momentum:
 $m_1 v_1 = m_2 v_2 \Rightarrow \frac{m_1}{m_2} = \frac{v_2}{v_1} = \frac{1}{8}$.
Radius $R \propto A^{1/3} \propto m^{1/3}$. Ratio $\frac{R_1}{R_2} = \left(\frac{1}{8}\right)^{1/3} = \frac{1}{2}$.
(NCERT Ed. 2025-26, 12th, Page No. 208)
3. (3)
From Einstein's equation $K_{\max} = h\nu - \phi$, the graph is a straight line ($y = mx + c$). The slope $m = h$ (Planck's constant), which is a universal constant independent of the metal. Ratio is 1:1.
(NCERT Ed. 2025-26, 12th, Page No. 187)
4. (2)
Width of secondary max $\approx \frac{D\lambda}{a}$.
 $W = \frac{1 \times 600 \times 10^{-9}}{0.2 \times 10^{-3}} = 3000 \times 10^{-6} \text{ m} = 3 \text{ mm}$.
(NCERT Ed. 2025-26, 12th, Page No. 204)
5. (4)
Energy of a state is given by $E_n \propto \frac{z^2}{n^2}$.
 $E_{4n} - E_{2n} \propto z^2 \left(\frac{1}{(2n)^2} - \frac{1}{(4n)^2} \right) = \frac{3z^2}{16n^2}$
 $E_{2n} - E_n \propto z^2 \left(\frac{1}{n^2} - \frac{1}{(2n)^2} \right) = \frac{3z^2}{4n^2}$ Ratio
 $= \frac{3/16}{3/4} = \frac{1}{4}$ (a constant). It does not depend on z or n .
(NCERT Ed. 2025-26, 12th, Page No. 159)
6. (2)
Radius, $R = R_0 A^{1/3} \Rightarrow R \propto A^{1/3}$
 $V \propto R^3$
 $\therefore V \propto A$
And mass $\propto A$
So, density is independent of A.
(NCERT Ed. 2025-26, 12th, Page No. 169)
7. (4)
Zener breakdown occurs in heavily doped junctions. Large doping makes the depletion layer very narrow, which creates a strong electric field across the junction. This field is strong enough to directly break covalent bonds and generate a large number of charge carriers
(NCERT Ed. 2025-26, 12th, Page No. 195)
8. (3)
Magnetic fields do not change the speed of an electron, so λ_2 stays constant. Electric fields exert a force that changes the speed, so λ_1 must increase or decrease depending on the direction. Thus, λ_1 will not be equal to λ_2 .
(NCERT Ed. 2025-26, 12th, Page No. 191)
9. (1)
The p-side (anode) is at -3V and the n-side (cathode) is connected toward ground (0V). Since $V_p < V_n$, the diode is **reverse biased**. No current flows through the circuit, meaning there is no voltage drop across the 5Ω resistor. Therefore, point A is at ground potential, 0V .
(NCERT Ed. 2025-26, 12th, Page No. 167)
10. (2)
Intensity $I = I_{\max} \cos^2 \left(\frac{\phi}{2} \right)$.
For $I = 0.75 I_{\max}$, phase difference $\phi = \frac{\pi}{3}$.
Using $\phi = \frac{2\pi}{\lambda} \cdot \frac{yd}{D}$,
We find the distance from central maxima
 $y = \frac{\lambda D}{6d}$. The distance between two such points on either side is
 $2y = \frac{\lambda D}{3d} = \frac{(600 \times 10^{-9}) \times 1}{3 \times 10^{-3}} = 2 \times 10^{-4} \text{ m}$
 $= 0.2 \text{ mm}$
(NCERT Ed. 2025-26, 12th, Page No. 158)
11. (2)
 $f = R/2 = +20 \text{ cm}$. Magnification $m = +1/3$ (virtual/erect).
 $m = \frac{f}{f-u} \Rightarrow \frac{1}{3} = \frac{20}{20-u} \Rightarrow 20-u = 60$
 $\Rightarrow u = -40 \text{ cm}$
(NCERT Ed. 2025-26, 12th, Page No. 208)

12. (3)
Maximum kinetic energy
 $= h\nu - \phi_0 = eV_s$
$$V_s = \frac{h\nu}{e} - \frac{\phi_0}{e}$$

(NCERT Ed. 2025-26, 12th, Page No. 196)

13. (4)
In Bohr's model, as the principal quantum number n increases (excited state):
Kinetic Energy ($K \propto \frac{1}{n^2}$) decreases.
Potential Energy ($U \propto -\frac{1}{n^2}$) becomes less negative, so it increases.
(NCERT Ed. 2025-26, 12th, Page No. 168)

14. (4)
When e^- jump from lower to higher energy level it absorbed energy.
Therefore statement I is wrong
For statement II
For the jumping of electron from higher energy orbit (E_0) to lower energy orbit (E_1)
 $h\nu = E_2 - E_1$
$$\nu = \frac{E_2 - E_1}{h}$$

(NCERT Ed. 2025-26, 12th, Page No. 162)

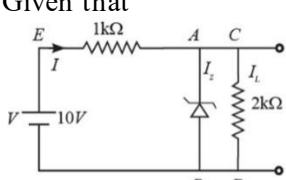
15. (2)
Energy levels $E_n \propto \frac{1}{n^2}$ and Bohr radius $r_n \propto n^2$.
If mass m doubles, all energies (including ionisation energy and gaps) double, while the radius is halved.
(NCERT Ed. 2025-26, 12th, Page No. 169)

16. (4)
Image by convex lens:
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{v} + \frac{1}{20} = \frac{1}{5}$$

$$v = \frac{20}{3} \text{ cm}$$

Image by concave lens: $u = \left(\frac{20}{3} - 2 \right) = \frac{14}{3} \text{ cm}$
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{v} - \frac{3}{14} = \frac{-1}{5}, v = 70 \text{ cm}$$

(NCERT Ed. 2025-26, 12th, Page No. 169)

17. (2)
Given that

 $V_Z = 3 \text{ V}$

Let potential at B = 0 V
Potential At E (V_E) = 10 V
 $V_C = V_A = 3 \text{ V}$
 $I_Z + I_1 = I$
 $I = V/R$

$$I = \frac{10-3}{1000} = \frac{7}{1000} \text{ A}$$

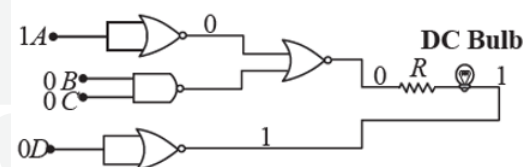
$$I_1 = \frac{3}{2000} \text{ A}$$

$$\text{Therefore } I_Z = \frac{7-1.5}{1000} = 5.5 \text{ mA}$$

(NCERT Ed. 2025-26, 12th, Page No. 160)

18. (1)
When a P - N junction diode is formed an electric field is developed from N -side to P -side due to which barrier potential is created & majority charge carrier can not flow through the junction due to barrier potential so current is zero unless we apply forward bias voltage.
(NCERT Ed. 2025-26, 12th, Page No. 168)

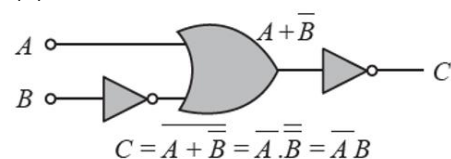
19. (2)
When bulb have potential drop on it, the bulb will glow.
One end of bulb must be at high (1) and other must be at low (0)



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20. (3)
• Emission of 1α particle decreases atomic number (Z) by 2.
• Emission of $1\beta^-$ particle increases atomic number (Z) by 1. If n α -particles are emitted, $\Delta Z = -2n$. To counter this, $2n$ β^- -particles must be emitted ($\Delta Z = +2n$). The net change in atomic number is zero, meaning the daughter has the same Z as the parent. Thus, it is an isotope.

(NCERT Ed. 2025-26, 12th, Page No. 191)

21. (4)

$$C = A + B = A \cdot B = \overline{A} \cdot \overline{B}$$

(NCERT Ed. 2025-26, 12th, Page No. 191)

22. (4)
• Step 1: Element A (${}_Z A^X$) undergoes α -decay to form B . The atomic number of B is $(Z - 2)$.

- Step 2: Element B (${}_{Z-2}^{X-4}B$) emits two β^- particles ($2e^-$) to form C . This increases the atomic number by 2, making the atomic number of C equal to $(Z-2)+2=Z$.

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23. (2)

In an n-type semiconductor, doping with **pentavalent** atoms (group 15) provides extra electrons. Thus, electrons become the majority carriers and **holes** become the minority carriers.

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24. (4)

Using $V = \frac{h}{e}(\nu - \nu_0)$, we have $V_1 = m(\nu_1 - \nu_0)$ and $V_2 = m(\nu_2 - \nu_0)$. Dividing these equations and solving for the threshold frequency ν_0 yields option 4.

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25. (4)

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV. For } He^+, Z = 2.$$

Ground state $n = 1$.

$$E = -13.6 \times \frac{2^2}{1^2} = -54.4 \text{ eV.}$$

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26. (4)

In reverse bias, the current is due to minority carriers and is very small. When light falls on the diode, the generated electron-hole pairs cause a large fractional change in the minority carrier concentration, making the change in current easily detectable.

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27. (3)

Nuclear radius $R \propto A^{1/3}$. Ratio

$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2} \right)^{1/3} = \left(\frac{125}{64} \right)^{1/3} = \frac{5}{4} = 1.25.$$

(NCERT Ed. 2025-26, 12th, Page No. 167)

28. (2)

Energy Released $Q = E_{\text{final}} - E_{\text{initial}}$ Since the total number of nucleons remains the same (200), we can use the difference in binding energy per nucleon:

$$Q = 200 \times (8.2 - 7.4) = 200 \times 0.8 = 160 \text{ MeV}$$

(NCERT Ed. 2025-26, 12th, Page No. 167)

29. (3)

If a mirror moves by a distance x , the image shifts by $2x$. Shift $= 2 \times 3 = 6 \text{ cm}$.

(NCERT Ed. 2025-26, 12th, Page No. 158)

30. (2)

We know that,

$$\beta = \frac{\lambda D}{d} \text{ and } \lambda_{\text{Red}} > \lambda_{\text{violet}}$$

Therefore, if the source of light used in a Young's double slit experiment is changed from red to violet fringe width decreases and fringe lines will come close.

(NCERT Ed. 2025-26, 12th, Page No. 204)

31. (2)

Object A ($u = 30, f = 20$): Real image.

$$v_A = \frac{uf}{u-f} = \frac{+600}{-10} = -60 \text{ cm.}$$

Object B ($u = 10, f = 20$) Virtual image.

$$v_B = \frac{+200}{10} = +20 \text{ cm.}$$

$$\text{Distance} = |v_A| + |v_B| = 60 + 20 = 80 \text{ cm}$$

(NCERT Ed. 2025-26, 12th, Page No. 187)

32. (4)

Angle of incidence $i = 60^\circ$. Deviation

$$\delta = i - r \Rightarrow 30^\circ = 60^\circ - r \Rightarrow r = 30^\circ$$

$$\mu = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

(NCERT Ed. 2025-26, 12th, Page No. 168)

33. (3)

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin(A/2)}. \text{ Given } A = 60^\circ.$$

$$\sqrt{2} = \frac{\sin(30^\circ + \delta_m/2)}{\sin 30^\circ}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \sin(30^\circ + \delta_m/2)$$

$$30^\circ + \delta_m/2 = 45^\circ \Rightarrow \delta_m = 30^\circ.$$

(NCERT Ed. 2025-26, 12th, Page No. 159)

34. (2)

For TIR, $i > C \Rightarrow \sin 45^\circ > \sin C$.

$$\frac{1}{\sqrt{2}} > \frac{1}{\mu} \Rightarrow \mu > \sqrt{2} \approx 1.414.$$

$$\mu = 1.5$$

(NCERT Ed. 2025-26, 12th, Page No. 169)

35. (2)

$$\tan C = \frac{R}{\text{depth}} = \frac{1}{\sqrt{3}} \Rightarrow C = 30^\circ.$$

(NCERT Ed. 2025-26, 12th, Page No. 195)

36. (2)

$$\text{Phase difference } \phi = \frac{2\pi}{\lambda} \left(\frac{\lambda}{4} \right) = \frac{\pi}{2}.$$

$$I = I_0 \cos^2 \left(\frac{\phi}{2} \right) = I_0 \cos^2 \left(\frac{\pi}{4} \right) = I_0 \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{I_0}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 191)

37. (3)

$$A_1 = 2, A_2 = 1.$$

$$A_{\max} = 2 + 1 = 3;$$

$$A_{\min} = 2 - 1 = 1.$$

$$I \propto A^2 \Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{3}{1} \right)^2 = \frac{9}{1}.$$

(NCERT Ed. 2025-26, 12th, Page No. 167)

38. (4)

For the fringe to be opposite a slit, $y = \frac{d}{2}$.

$$\text{For 1st dark fringe, } y = \frac{D\lambda}{2d}.$$

$$\frac{d}{2} = \frac{D\lambda}{2d} \Rightarrow \lambda = \frac{d^2}{D} = \frac{(5 \times 10^{-4})^2}{0.5} = 500 \times 10^{-9} \text{ m}$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

39. (2)

$$\text{Kinetic Energy } K = |\text{Total Energy}| = 1.51 \text{ eV}.$$

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA} = \frac{12.27}{\sqrt{1.51}} \approx \frac{12.27}{1.23} \approx 10 \text{ \AA}.$$

(NCERT Ed. 2025-26, 12th, Page No. 168)

40. (3)

$$I_{\text{out}} = \frac{I_0}{8} \sin^2(2\theta).$$

$$1 = \frac{32}{8} \sin^2(2\theta) \Rightarrow \sin^2(2\theta)$$

$$= \frac{1}{4} \Rightarrow \sin(2\theta) = \frac{1}{2}$$

$$2\theta = 30^\circ \Rightarrow \theta = 15^\circ.$$

(NCERT Ed. 2025-26, 12th, Page No. 196)

41. (2)

$$\frac{hc}{\lambda} = \phi + 3eV \dots (i) \quad \frac{hc}{2\lambda} = \phi + 1eV \dots (ii) \text{ From (ii)}$$

$$\frac{hc}{\lambda} = 2\phi + 2eV. \text{ Equating with (i):}$$

$\phi + 3eV = 2\phi + 2eV \Rightarrow \phi = eV$. Substitute into (i):

$$\frac{hc}{\lambda} = 4eV = 4\phi. \quad \lambda_{th} = \frac{hc}{\phi} = 4\lambda.$$

(NCERT Ed. 2025-26, 12th, Page No. 166)

42. (2)

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

$$\text{For } 3 \rightarrow 1: \frac{1}{\lambda_1} \propto \left(1 - \frac{1}{9} \right) = \frac{8}{9}.$$

$$\text{For } 2 \rightarrow 1: \frac{1}{\lambda_2} \propto \left(1 - \frac{1}{4} \right) = \frac{3}{4}.$$

$$\frac{\lambda_1}{\lambda_2} = \frac{3/4}{8/9} = \frac{27}{32}.$$

(NCERT Ed. 2025-26, 12th, Page No. 162)

43. (4)

Semiconductors have a negative temperature coefficient of resistance. As temperature increases, more covalent bonds break, generating more electron-hole pairs, which increases conductivity and thus decreases resistance.

(NCERT Ed. 2025-26, 12th, Page No. 167)

44. (3)

$$E = mc^2 \Rightarrow P = \frac{m}{t} c^2.$$

$$\frac{m}{t} = \frac{P}{c^2} = \frac{900 \times 10^6}{(3 \times 10^8)^2} = 10^{-8} \text{ kg/s. Mass per hour}$$

$$= 10^{-8} \times 3600 = 36 \times 10^{-6} \text{ kg} = 36 \text{ mg}$$

(NCERT Ed. 2025-26, 12th, Page No. 169)

45. (2)

Using Einstein's photoelectric equation,

$$K = \frac{hc}{\lambda} - \phi.$$

$$\text{Case 1: } K_1 = \frac{hc}{\lambda} - \phi$$

$$\text{Case 2: } K_2 = \frac{2hc}{\lambda} - \phi \text{ Given}$$

$$K_2 = 3K_1 \Rightarrow \frac{2hc}{\lambda} - \phi = 3 \left(\frac{hc}{\lambda} - \phi \right)$$

$$\Rightarrow \frac{2hc}{\lambda} - \phi = \frac{3hc}{\lambda} - 3\phi \Rightarrow 2\phi = \frac{hc}{\lambda} \Rightarrow \phi = \frac{hc}{2\lambda}$$

(NCERT Ed. 2025-26, 12th, Page No. 160)

46. (1)
AgNO₃ precipitates only ionisable (outer-sphere) Cl⁻.
 • outer Cl⁻ = 3 → 3 AgCl
 • outer Cl⁻ = 2 → 2 AgCl
 • outer Cl⁻ = 1 → 1 AgCl
 (NCERT Ed. 2025-26, 12th, Page No. 119)
47. (2)
From the table (kJ mol⁻¹): Ti = 473, V = 515, Cr = 397, Zn = 126. Highest is V (515).
 (NCERT Ed. 2025-26, 12th, Page No. 99)
48. (4)
Densities: Al 2.70, Ga 5.90, In 7.31, Tl 11.85 → highest is Tl.
 (NCERT Ed. 2022-23, 11th, Page No. 318)
49. (3)
Type is [MABCD] (all different) in square planar → 3 geometrical isomers.
 (NCERT Ed. 2025-26, 12th, Page No. 125)
50. (1)
From the table (V): Ti = -1.63, Fe = -0.44, Cu = +0.34, Zn = -0.76.
 Most negative = Ti (-1.63).
 (NCERT Ed. 2025-26, 12th, Page No. 99)
51. (3)
Melting points (K): B 2453, Al 933, Ga 303, In 430 → lowest is Ga (303 K).
 (NCERT Ed. 2022-23, 11th, Page No. 318)
52. (1)
Covalent radii (pm): C 77, Si 118, Ge 122, Sn 140, Pb 146 → increasing order is C < Si < Ge < Sn < Pb.
 (NCERT Ed. 2022-23, 11th, Page No. 323)
53. (2)
cis-[Co(en)₂Cl₂]⁺ is chiral → forms non-superimposable mirror images.
 trans-[Co(en)₂Cl₂]⁺ has symmetry → optically inactive.
 (NCERT Ed. 2025-26, 12th, Page No. 126)
54. (3)
M²⁺ radii (pm): Fe²⁺ = 77, Co²⁺ = 74, Ni²⁺ = 70, Cu²⁺ = 73.
 Smallest = Ni²⁺ (70 pm).
 (NCERT Ed. 2025-26, 12th, Page No. 94)
55. (3)
Given text: iron catalysts are used in the Haber process.
 (NCERT Ed. 2022-23, 12th, Page No. 176)
56. (4)
Δ_iH₁ (kJ/mol): C 1086, Si 786, Ge 761, Pb 715 → highest is C.
 (NCERT Ed. 2022-23, 11th, Page No. 323)
57. (1)
 • [Fe(CN)₆]⁴⁻: Fe²⁺ = d⁶, CN⁻ strong field → low spin, 0 unpaired → 0 BM
 • [FeF₆]³⁻: Fe³⁺ = d⁵, F⁻ weak field → high spin, 5 unpaired → 5.92 BM
 • [Fe(H₂O)₆]²⁺: Fe²⁺ = d⁶, H₂O weak → high spin, 4 unpaired → 4.90 BM
 • [Fe(CN)₆]³⁻: Fe³⁺ = d⁵, CN⁻ strong → low spin, 1 unpaired → 1.73 BM
 (NCERT Ed. 2025-26, 12th, Page No. 130)
58. (3)
Mn₂O₇ has Mn in +7, the highest in the table.
 (NCERT Ed. 2025-26, 12th, Page No. 100)
59. (1)
From table: I₁N 1402 > P 1012 > As 947 > Sb 834 > Bi 703 (decreases). Reason matches.
 (NCERT Ed. 2022-23, 12th, Page No. 171)
60. (2)
Spectrochemical series (relevant part):
 F⁻ < H₂O < NH₃ < CN⁻
 (NCERT Ed. 2025-26, 12th, Page No. 132)
61. (2)
Oxygen stabilises high oxidation states strongly due to its ability to form multiple bonds and structures like Mn–O–Mn bridges, etc.
 (NCERT Ed. 2025-26, 12th, Page No. 100)

62. (3)
Table explicitly shows $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$.
(NCERT Ed. 2022-23, 12th, Page No. 177)
63. (3)
 $\Delta_t \approx \frac{4}{9}\Delta_0$
 $\Delta_t = \frac{4}{9} \times 18000 = 8000 \text{ cm}^{-1}$
(NCERT Ed. 2025-26, 12th, Page No. 133)
64. (1)
Table: H_3PO_3 is +3 and $\text{H}_4\text{P}_2\text{O}_5$ is +3 (same).
(NCERT Ed. 2022-23, 12th, Page No. 184)
65. (2)
Table BDE: F_2 158.8, Cl_2 242.6, Br_2 192.8, I_2 151.1
→ maximum is Cl_2 .
(NCERT Ed. 2022-23, 12th, Page No. 198)
66. (3)
Given example hydrolysis:
 $2\text{XeF}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
So products: Xe, HF, O_2 .
(NCERT Ed. 2022-23, 12th, Page No. 210)
67. (3)
Octahedral $[\text{M}(\text{en})_2\text{X}_2]^+$ has cis and trans.
 - cis form is optically active → gives 2 enantiomers
 - trans form is optically inactive → gives 1
Total = 2 + 1 = 3
(NCERT Ed. 2025-26, 12th, Page No. 126)
68. (2)
In presence of iodide, Cu^{2+} oxidises I^- to I_2 and gets reduced to Cu^+ forming CuI :
 $2\text{Cu}^{2+} + 4\text{I}^- \rightarrow 2\text{CuI(s)} + \text{I}_2$.
(NCERT Ed. 2025-26, 12th, Page No. 100)
69. (1)
In acidic water, Al^{3+} exists mainly as hexaaqua complex
 $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ (octahedral) → hybridisation sp^3d^2 .
(NCERT Ed. 2022-23, 11th, Page No. 319)

70. (3)
The given text explicitly states: hydrolysis of XeF_6 does not lead to a redox reaction; oxidation states remain same as in reactants.
(NCERT Ed. 2022-23, 12th, Page No. 211)
71. (2)
Lanthanoid contraction makes 5d elements (like Hf) smaller than expected, making Hf radius very similar to Zr.
(NCERT Ed. 2025-26, 12th, Page No. 109)
72. (3)
 - $\text{XeF}_2 + \text{PF}_5 \rightarrow [\text{XeF}]^+[\text{PF}_6]^-$ (not $[\text{XeF}]^{+?}$
Actually yes, cation is $[\text{XeF}]^+$)
 - $\text{XeF}_4 + \text{SbF}_5 \rightarrow [\text{XeF}_3]^+[\text{SbF}_6]^-$
 - $\text{XeF}_6 + \text{MF} \rightarrow \text{M}^+[\text{XeF}_7]^-$ where M = Na, K, Rb, Cs
Option (3) matches exactly and is the most direct from text.
(NCERT Ed. 2022-23, 12th, Page No. 210)
73. (2)
 $\lambda \propto \frac{1}{\Delta_0}$. Stronger ligand $\Rightarrow$ larger $\Delta_0 \Rightarrow$ shorter λ .
Strength: $\text{F}^- < \text{H}_2\text{O} < \text{NH}_3$
So $\Delta_0 : \text{F}^-$ smallest $\Rightarrow \lambda$ largest
Increasing λ : NH_3 (smallest λ) $< \text{H}_2\text{O} < \text{F}^-$ (largest λ)
(NCERT Ed. 2025-26, 12th, Page No. 133)
74. (3)
Colour in transition metal ions is mainly due to d-d transitions.
 - Zn^{2+} is $3\text{d}^{10} \rightarrow$ no d-d transition $\rightarrow$ colourless (as per table).
(NCERT Ed. 2025-26, 12th, Page No. 103)
75. (3)
Brown ring complex = nitrosyl ferrous sulphate
 $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$.
(NCERT Ed. 2022-23, 12th, Page No. 180)
76. (3)
 Ce^{3+} has $4\text{f}^1 \rightarrow$ one unpaired electron $\rightarrow$ paramagnetic.
 La^{3+} (f^0) and Lu^{3+} (f^{14}) are not paramagnetic.
(NCERT Ed. 2025-26, 12th, Page No. 110)

77. (1)
Zero CFSE for high-spin d^5 (also d^0 , d^{10}).
 $Mn^{2+} = d^5$ and H_2O is weak field $\Rightarrow$ high spin $d^5 \Rightarrow$ CFSE = 0.
Others: d^8 , d^6 , d^6 (low spin possible) $\Rightarrow$ CFSE $\neq 0$.
(NCERT Ed. 2025-26, 12th, Page No. 132)
78. (1)
 S^{2-} gives H_2S (rotten egg) and forms black PbS.
(NCERT Ed. 2025-26 12th Part-I Page No. 00)
79. (3)
The text states La^{3+} and Lu^{3+} do not show colour in aqueous solution (f^0 / f^{14}).
(NCERT Ed. 2025-26, 12th, Page No. 110)
80. (3)
 - Inner-orbital generally implies pairing due to strong field ligand (for d^4 – d^7 octahedral) or dsp^2 square planar.
 - $Fe^{2+} = d^6$, H_2O weak field $\rightarrow$ high spin (unpaired) $\rightarrow$ paramagnetic, outer orbital.
 - $Co^{3+} = d^6$, F^- weak field $\rightarrow$ high spin $\rightarrow$ paramagnetic.
 - $[Ni(CN)_4]^{2-}$: $Ni^{2+} = d^8$, CN^- strong field $\rightarrow$ square planar (dsp^2) $\rightarrow$ all paired $\rightarrow$ diamagnetic and inner-orbital.
 - $Zn^{2+} = d^{10}$, diamagnetic yes, but tetrahedral/square planar depends; typically $[Zn(NH_3)_4]^{2+}$ is tetrahedral (sp^3) $\rightarrow$ outer orbital conceptually.
(NCERT Ed. 2025-26, 12th, Page No. 130)
81. (1)
 - $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$
 - $Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$
(NCERT Ed. 2025-26, 12th, Page No. 106)
82. (3)
Group II (acidic H_2S) includes Pb^{2+} , Cu^{2+} , Cd^{2+} , Hg^{2+} , Bi^{3+} , As/Sb/Sn
(NCERT Ed. 2025-26 12th Part-I Page No. 00)
83. (2)
 NH_4^+ gives NH_3 on heating with base; NH_3 forms NH_4Cl fumes with HCl.
(NCERT Ed. 2025-26 12th Part-I Page No. 00)
84. (2)
 - In $[Ni(CN)_4]^{2-}$: CN^- strong field $\rightarrow$ pairing $\rightarrow$ square planar $\rightarrow dsp^2$.
 - In $[NiCl_4]^{2-}$: Cl^- weak field $\rightarrow$ no extra pairing $\rightarrow$ tetrahedral $\rightarrow sp^3$.
(NCERT Ed. 2025-26, 12th, Page No. 130)
85. (3)
Lanthanoids predominantly form Ln^{3+} ions; +2 and +4 occur only for some elements.
(NCERT Ed. 2025-26, 12th, Page No. 109)
86. (1)
 SO_4^{2-} gives a white precipitate of $BaSO_4$ with $BaCl_2$ in acidic medium:
 $Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4 \downarrow$ (white) $BaSO_4$ is highly insoluble in dilute HCl and other dilute acids; hence it does not dissolve. $BaSO_4$ has very low solubility product and remains insoluble in water and dilute acids. Acidification with dilute HCl removes CO_3^{2-} interference by converting it into CO_2 gas, preventing formation of $BaCO_3$ precipitate, ensuring selective precipitation of $BaSO_4$.
87. (2)
Nitrite gives $HNO_2 \rightarrow$ decomposes to NO/NO_2 (brown fumes). It oxidizes iodide to iodine $\rightarrow$ blue with starch.
(NCERT Ed. 2025-26 12th Part-I Page No. 00)
88. (3)
 - Each en ligand is bidentate $\rightarrow$ contributes 2 donor atoms.
So, $2 \times en = 2 \times 2 = 4$ coordination sites.
 - Plus $2 \times Cl^- = 2$ sites.
Total coordination number = $4 + 2 = 6$.
(NCERT Ed. 2025-26, 12th, Page No. 121)
89. (2)
Manganate ion and permanganate ion are tetrahedral. Manganate is paramagnetic (one unpaired electron)
(NCERT Ed. 2025-26, 12th, Page No. 107)
90. (2)
From the given data: In neutral/faintly alkaline solution, a notable reaction is oxidation of iodide to iodate:
 $2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$
So medium must be neutral or faintly alkaline.
(NCERT Ed. 2025-26, 12th, Page No. 108)

91. (4)

A population has certain attributes whereas, an individual organism does not. An individual may have births and deaths, but a population has birth rates and death rates. Another attribute characteristic of a population is sex ratio.

(NCERT Ed. 2025-26, 12th, Page No. 191)

92. (1)

Auxins induce parthenocarpy, e.g., in tomatoes. Spraying juvenile conifers with GAs hastens the maturity period, thus leading to early seed production. Cytokinins promote nutrient mobilisation which helps in the delay of leaf senescence. Ethylene promotes female flowers in cucumbers thereby increasing the yield.

(NCERT Ed. 2025-26, 11th, Page No. 176)

93. (3)

Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

(NCERT Ed. 2025-26, 12th, Page No. 199)

94. (2)

The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth- it is home to more than 40,000 species of plants, 3,000 of fishes, 1,300 of birds, 427 of mammals, 427 of amphibians, 378 of reptiles and of more than 1,25,000 invertebrates.

(NCERT Ed. 2025-26, 12th, Page No. 219)

95. (4)

Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in build up of organic materials. Decomposition rate is slower if detritus is rich in lignin and chitin, and quicker, if detritus is rich in nitrogen and water-soluble substances like sugars.

(NCERT Ed. 2025-26, 12th, Page No. 208)

96. (2)

The exponential growth can be expressed as $W_t = W_0 e^{rt}$. Here, r is the relative growth rate and is also the measure of the ability of the plant to produce new plant material, referred to as efficiency index.

In arithmetic growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures. In geometrical growth, both the progeny cells following mitotic cell division retain the ability to divide and continue to do so.

(NCERT Ed. 2025-26, 11th, Page No. 170)

97. (3)

Ramdeo Misra was honoured with the Fellowships of the Indian National Science Academy and World Academy of Arts and Science, and the prestigious Sanjay Gandhi Award in Environment and Ecology.

(NCERT Ed. 2025-26, 12th, Page No. 189, 190)

98. (1)

The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceeds that of phytoplankton.

(NCERT Ed. 2025-26, 12th, Page No. 213)

99. (3)

Three of the hotspots – Western Ghats and Sri Lanka, Indo-Burma and Himalaya – cover our country's exceptionally high biodiversity regions.

(NCERT Ed. 2025-26, 12th, Page No. 224)

100. (2)

The number of trophic levels in the grazing food chain is restricted as the transfer of energy follows 10 per cent law – only 10 per cent of the energy is transferred to each trophic level from the lower trophic level.

- Producer (P): 10,000 J
- Primary Consumer (PC): 10% of 10,000 J = 1000 J
- Secondary Consumer (SC): 10% of 1000 J = 100 J

(NCERT Ed. 2025-26, 12th, Page No. 211)

101. (3)

The 'bakanae' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen *Gibberella fujikuroi*. E. Kurosawa (1926) reported the appearance of symptoms of the disease in rice seedlings when they were treated with sterile filtrates of the fungus.

(NCERT Ed. 2025-26, 11th, Page No. 175)

102. (4)

Both the species benefit in mutualism and both lose in competition in their interactions with each other. The interaction where one species is benefitted and the other is neither benefitted nor harmed is called commensalism. In amensalism on the other hand one species is harmed whereas the other is unaffected.

(NCERT Ed. 2025-26, 12th, Page No. 197)

103. (1)

Many species extinctions in the last 500 years (Steller's sea cow, passenger pigeon) were due to overexploitation by humans.

(NCERT Ed. 2025-26, 12th, Page No. 223)

104. (2)

Of the incident solar radiation less than 50 per cent of it is photosynthetically active radiation (PAR).

(NCERT Ed. 2025-26, 12th, Page No. 209)

105. (3)

In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called apical dominance. Removal of shoot tips (decapitation) usually results in the growth of lateral buds. It is widely applied in tea plantations and hedge-making.

(NCERT Ed. 2025-26, 11th, Page No. 175)

106. (2)

Saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

(NCERT Ed. 2025-26, 12th, Page No. 214)

107. (2)

From the given data, a spindle-shaped age pyramid is formed which indicates declining population.

(NCERT Ed. 2025-26, 12th, Page No. 191)

108. (4)

During his pioneering and extensive explorations in the wilderness of South American jungles, the great German naturalist and geographer Alexander von Humboldt observed that within a region species richness increased with increasing explored area, but only up to a limit. In fact, the relation between species richness and area for a wide variety of taxa

(angiosperm plants, birds, bats, freshwater fishes) turns out to be a rectangular hyperbola. On a logarithmic scale, the relationship is a straight line.

(NCERT Ed. 2025-26, 12th, Page No. 220)

109. (2)

Auxins like IAA and indole butyric acid (IBA) have been isolated from plants. NAA (naphthalene acetic acid) and 2, 4-D (2, 4- dichlorophenoxyacetic) are synthetic auxins.

(NCERT Ed. 2025-26, 11th, Page No. 175)

110. (1)

Some examples of recent extinctions include the dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger.

(NCERT Ed. 2025-26, 12th, Page No. 221)

111. (2)

Sugarcane stores carbohydrate as sugar in their stems. Spraying sugarcane crop with gibberellins increases the length of the stem, thus increasing the yield by as much as 20 tonnes per acre.

(NCERT Ed. 2025-26, 11th, Page No. 176)

112. (4)

Forest, grassland and desert are some examples of terrestrial ecosystems; pond, lake, wetland, river and estuary are some examples of aquatic ecosystems. Crop fields and an aquarium may also be considered as man-made ecosystems.

(NCERT Ed. 2025-26, 12th, Page No. 207)

113. (1)

If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food. This is the reason why predators in nature are 'prudent'. Prey species have evolved various defenses to lessen the impact of predation. Some species of insects and frogs are cryptically-coloured (camouflaged). Some are poisonous and therefore avoided by the predators.

(NCERT Ed. 2025-26, 12th, Page No. 198)

114. (2)

Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity, e.g., heterophylly in cotton, coriander and larkspur.

(NCERT Ed. 2025-26, 11th, Page No. 173)

115. (3)

In a pond ecosystem, the autotrophic components include the phytoplankton, some algae and the floating, submerged and marginal plants found at the edges. The consumers are represented by the zooplankton, the free swimming and bottom dwelling forms. The decomposers are the fungi, bacteria and flagellates especially abundant in the bottom of the pond.

(NCERT Ed. 2025-26, 12th, Page No. 206)

116. (4)

The magnitude of r value, for the Norway rat is 0.015, and for the flour beetle it is 0.12. In 1981, the r value for human population in India was 0.0205.

(NCERT Ed. 2025-26, 12th, Page No. 194)

117. (1)

ABA stimulates the closure of stomata and increases the tolerance of plants to various kinds of stresses.

(NCERT Ed. 2025-26, 11th, Page No. 177)

118. (2)

According to the International Union for Conservation of Nature and Natural Resources (IUCN) (2004), the total number of plant and animal species described so far is slightly more than 1.5 million. A more conservative and scientifically sound estimate made by Robert May places the global species diversity at about 7 million. More than 70 per cent of all the species recorded are animals, while plants comprise no more than 22 per cent of the total. The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.

(NCERT Ed. 2025-26, 12th, Page No. 217)

119. (3)

Algae are one of the producers in aquatic ecosystems.

(NCERT Ed. 2025-26, 12th, Page No. 206)

120. (3)

Populations evolve to maximise their reproductive fitness, also called Darwinian fitness (high r value), in the habitat in which they live. Under a particular set of selection pressures, organisms evolve towards the most efficient reproductive strategy. Some organisms breed only once in their lifetime (Pacific salmon fish, bamboo) while others breed many times during their lifetime (most birds and mammals).

(NCERT Ed. 2025-26, 12th, Page No. 196)

121. (4)

In arithmetic growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures. The simplest expression of arithmetic growth is exemplified by a root elongating at a constant rate. On plotting the length of the organ against time, a linear curve is obtained.

(NCERT Ed. 2025-26, 11th, Page No. 170)

122. (3)

Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable.

(NCERT Ed. 2025-26, 12th, Page No. 220)

123. (2)

Humification leads to accumulation of a dark coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate. Being colloidal in nature it serves as a reservoir of nutrients. The humus is further degraded by some microbes and release of inorganic nutrients occur by the process known as mineralisation.

(NCERT Ed. 2025-26, 12th, Page No. 208)

124. (4)

When we think of predator and prey, most probably it is the tiger and the deer that readily come to our mind, but a sparrow eating any seed is no less a predator. Ticks on dogs are parasites. In some shallow South American lakes, visiting flamingoes and resident fishes compete for their common food, the zooplankton in the lake. Barnacles growing on the back of a whale benefits, but the whale does not derive any apparent benefit from this interaction i.e. commensalism.

(NCERT Ed. 2025-26, 12th, Page No. 198, 200, 201)

125. (3)

Secondary productivity is defined as the rate of formation of new organic matter by consumers.

(NCERT Ed. 2025-26, 12th, Page No. 207)

126. (1)

If 4 individuals in a laboratory population of 40 fruitflies died during a specified time interval, say a week, the death rate in the population during that period is $4/40 = 0.1$ individuals per fruitfly per week.

(NCERT Ed. 2025-26, 12th, Page No. 191)

127. (4)

In most situations, ABA acts as an antagonist to GAs.

(NCERT Ed. 2025-26, 11th, Page No. 178)

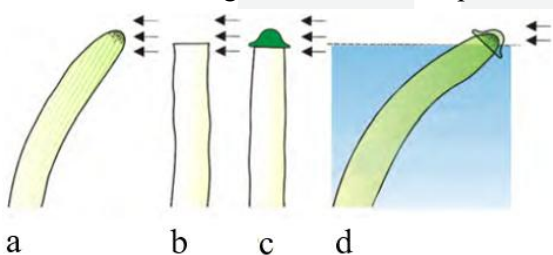
128. (2)

Tilman found that plots with more species showed less year-to-year variation in total biomass. He also showed that in his experiments, increased diversity contributed to higher productivity. MacArthur showed that five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities.

(NCERT Ed. 2025-26, 12th, Page No. 221)

129. (3)

Charles Darwin and his son Francis Darwin observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism). After a series of experiments, it was concluded that the tip of coleoptile was the site of transmittable influence that caused the bending of the entire coleoptile.



Experiment used to demonstrate that tip of the coleoptile is the source of auxin. Arrows indicate direction of light

(NCERT Ed. 2025-26, 11th, Page No. 174)

130. (1)

Although the term interbreeding implies sexual reproduction, a group of individuals resulting from even asexual reproduction is also generally considered a population for the purpose of ecological studies. Although an individual organism is the one that has to cope with a changed environment, it is at the population level that natural selection operates to evolve the desired traits.

(NCERT Ed. 2025-26, 12th, Page No. 191)

131. (3)

When there are situations where an animal or plant is endangered or threatened and needs urgent

measures to save it from extinction, *ex situ* (off site) conservation is the desirable approach.

(NCERT Ed. 2025-26, 12th, Page No. 224)

132. (1)

Species composition and stratification are the two main structural features of an ecosystem. Productivity, decomposition, energy flow, and nutrient cycling are the four important components of an ecosystem.

(NCERT Ed. 2025-26, 12th, Page No. 206)

133. (4)

The living differentiated cells, that by now have lost the capacity to divide can regain the capacity of division under certain conditions. This phenomenon is termed as dedifferentiation. For example, formation of meristems – interfascicular cambium and cork cambium from fully differentiated parenchyma cells.

(NCERT Ed. 2025-26, 11th, Page No. 172)

134. (2)

Under normal conditions, births and deaths are the most important factors influencing population density, the other two factors assuming importance only under special conditions. For instance, if a new habitat is just being colonised, immigration may contribute more significantly to population growth than birth rates.

(NCERT Ed. 2025-26, 12th, Page No. 193)

135. (3)

Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation. By the process of leaching, water-soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as catabolism. Humification leads to accumulation of a dark coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate.

(NCERT Ed. 2025-26, 12th, Page No. 207)

136. (3)

The innate immune system provides the first line of defense through physical barriers and non-specific responses. Skin prevents microorganism entry, lysozyme in tears and saliva destroys bacterial cell walls, and stomach acid denatures pathogens. All these are innate immune functions. However, the production of specific antibodies against particular pathogens is an adaptive immune system function, not innate. The adaptive system develops slowly and produces targeted, specific responses to pathogens previously encountered.

(NCERT Ed. 2025-26, 12th, Page No. 47)

137. (1)

List-I	List-II
Cotton bollworm	<i>CryIAc</i> gene
Corn borer	<i>CryIAb</i> gene
Tobacco infected with nematodes	dsRNA interference
<i>Bacillus thuringiensis</i>	Produce insecticidal protein crystals during a particular phase of their growth

(NCERT Ed. 2025-26, 12th, Page No. 48)

138. (4)

Restriction enzymes do not always cut DNA to produce blunt ends. They recognize palindromic sequences and cut both strands at specific distances from the center, often producing sticky ends (overhanging sequences) instead of blunt ends. For example, *EcoRI* cuts GAATTC producing sticky ends. Restriction enzymes do recognize palindromic sequences and cut at specific distances.

(NCERT Ed. 2025-26, 12th, Page No. 121)

139. (4)

- HIV is a retrovirus with RNA genome.
- It preferentially attacks CD₄⁺ helper T-lymphocytes, causing immunodeficiency.
- The ELISA diagnostic test detects HIV antigens or antibodies through antigen-antibody interaction.
- However, HIV is not transmitted through saliva and sweat—it requires direct contact with blood, sexual fluids, or mother-to-child transmission.
- Anti-retroviral drugs cannot completely cure AIDS—they suppress viral replication but don't eliminate the virus.

(NCERT Ed. 2025-26, 12th, Page No. 44)

140. (2)

After successfully cloning the insulin A and B chains separately in *E. coli*, the major technical challenge is creating the correct disulfide bonds between A and B chains. Simply producing the chains independently isn't sufficient; they must be properly joined with correct disulfide linkages to form functional insulin. This requires careful protein engineering, purification, oxidation conditions, and quality control to ensure the three-dimensional structure necessary for insulin activity.

(NCERT Ed. 2025-26, 12th, Page No. 122)

141. (3)

- Macrophages and neutrophils are phagocytic cells belonging to the innate immune system, not the adaptive system. They provide non-specific cellular responses—meaning they attack any foreign pathogen without prior exposure. They do not produce specific antibodies; that is the function of B-lymphocytes in the adaptive system.
- The innate immune system provides the first line of defense with non-specific responses.

(NCERT Ed. 2025-26, 12th, Page No. 46)

142. (4)

PCR (Polymerase Chain Reaction) requires three essential components: *Taq* polymerase (heat-stable DNA polymerase), primers (short DNA sequences complementary to the target DNA), and deoxyribonucleotides (dNTPs—building blocks for new DNA synthesis). These three are absolutely necessary for DNA amplification. However, restriction enzymes are not needed for PCR—they are used to cut DNA, not amplify it. Restriction enzymes are used after PCR if needed for cloning or analysis.

(NCERT Ed. 2025-26, 12th, Page No. 111)

143. (4)

List-I	List-II
pBR322	Plasmid with amp ^R and tet ^R genes
Restriction enzymes	Molecular scissors
Ti plasmid	From <i>Agrobacterium tumefaciens</i>
DNA ligase	Molecular glue

(NCERT Ed. 2025-26, 12th, Page No. 118)

144. (3)

Malaria (*Plasmodium falciparum*—female *Anopheles* mosquito), Typhoid (*Salmonella typhi*—contaminated food/water), and Amoebiasis (*Entamoeba histolytica*—large intestine) are all correctly paired. However, Filariasis caused by *Wuchereria bancrofti* is transmitted by female *Culex* mosquitoes or female *Anopheles* mosquitoes in some regions. The sand fly transmits other diseases like leishmaniasis and kala-azar.

(NCERT Ed. 2025-26, 12th, Page No. 113)

145. (3)

- Totipotency allows a single plant cell to develop into an entire plant.
- Micropropagation produces genetically identical somaclones (vegetative clones). explants must be cultured in nutrient medium containing minerals, vitamins, and plant hormones (auxins and cytokinins)—not just mineral salts alone.
- Meristem culture produces virus-free plants.
- Somatic hybridization requires protoplast fusion.

(NCERT Ed. 2025-26, 12th, Page No. 44)

146. (3)

The thymus gland is very large at birth and during childhood but reduces in size after puberty because T-lymphocytes have already developed sufficiently. The thymus produces thymosins (hormones) that promote T-cell maturation and development of cell-mediated immunity. B-lymphocytes mature in bone marrow.

(NCERT Ed. 2025-26, 12th, Page No. 122)

147. (3)

Bt toxin is selective and active only in insects with appropriate midgut pH and specific receptors (lepidopterans and some coleopterans)—not all insects. The toxin exists as inactive protoxin in *Bacillus thuringiensis* but becomes active in the insect's alkaline gut environment. *Bt* toxin binds to midgut epithelial cells and forms pores in the intestinal wall, disrupting digestion and causing insect death. This is the correct mechanism of action.

(NCERT Ed. 2025-26, 12th, Page No. 45)

148. (3)

List-I (Pathogenic organism)	List-II (Mode of transmission)
<i>Trichophyton</i>	Soil and contaminated clothes
Rhino virus	Droplets from infected person
<i>Wuchereria bancrofti</i>	Female <i>Culex</i> mosquito
<i>Entamoeba histolytica</i>	Housefly as mechanical carriers

(NCERT Ed. 2025-26, 12th, Page No. 48)

149. (1)

- Antibodies are produced by B-lymphocytes.
- Cell-mediated immunity is provided by T-lymphocytes.
- IgG antibodies provide long-term immunity—they are most abundant in blood and provide lasting protection after initial sensitization.
- Bone marrow is the production site for all blood cells, including lymphocytes that originate here before differentiating elsewhere.

(NCERT Ed. 2025-26, 12th, Page No. 122)

150. (4)

- DNA ligase creates phosphodiester bonds between DNA strands.
- Competent bacteria treated with CaCl₂ become permeable to plasmid uptake.
- Gel electrophoresis separates DNA fragments based on molecular weight and size.
- However, bioreactors maintain optimal conditions for both bacterial and mammalian cell cultures, not just bacteria. They control temperature, pH, dissolved oxygen, and nutrient levels for various culture types.

(NCERT Ed. 2025-26, 12th, Page No. 47)

151. (2)

Adaptive immunity doesn't provide immediate protection upon first exposure—that's why initial infections can be severe. However, adaptive immunity generates memory B cells and memory T cells that quickly recognize pathogens encountered previously. These memory cells allow secondary immune response to be faster and stronger than primary response. This is why vaccination protects us and why we develop immunity to childhood diseases like chickenpox.

(NCERT Ed. 2025-26, 12th, Page No. 121)

152. (1)

EcoRI restriction enzyme cuts the palindromic sequence GAATTC and produces sticky ends (overhanging sequences).

Taq polymerase synthesizes DNA at high temperatures but doesn't denature DNA (it works optimally at 72°C for extension).

DNA ligase joins DNA ends to create phosphodiester bonds; it doesn't remove nucleotides. These enzyme functions are essential for recombinant DNA technology.

Reverse transcriptase enzyme synthesizes double-stranded DNA from single-stranded RNA template.

(NCERT Ed. 2025-26, 12th, Page No. 119)

153. (4)

List-I	List-II
Bioprocess engineering	Maintenance of sterile ambience in chemical engineering processes
PCR	Detecting HIV in early asymptomatic stage
ELISA	Determining presence of antigens or antibodies
Blue-white screening	Identifying insertional inactivation

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154. (4)

ADA deficiency requires long-term management. The most appropriate treatment is gene therapy introducing the functional ADA gene into stem cells at early stages, particularly hematopoietic stem cells in bone marrow. This allows continuous ADA production from self-renewing stem cells throughout life. While enzyme replacement therapy provides temporary relief, only gene therapy offers lasting improvement. Bone marrow transplantation alone has limitations, and germline therapy raises ethical concerns.

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155. (4)

- Cancer cells lose contact inhibition, allowing unlimited growth in multiple layers.
- Cancer cells undergo unlimited division, escaping cellular senescence.
- Malignant tumors show metastasis—the ability to spread to distant sites. However, benign tumors do not metastasize; metastasis distinguishes malignant from benign tumors.

- Proto-oncogenes are normal growth-promoting genes; mutations convert them to oncogenes.

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156. (4)

DNA moves toward the positive electrode due to negative charges on its phosphate backbone. Smaller DNA fragments migrate faster through gel pores than larger fragments. Agarose gel provides a sieving effect based on pore size, separating molecules by size. However, ethidium bromide intercalation doesn't stop DNA movement—it intercalates between DNA base pairs and makes DNA fluorescent under UV light for visualization. DNA continues moving during electrophoresis after intercalation.

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157. (2)

The origin of replication (*ori*) is the DNA sequence where replication begins. For a plasmid to replicate autonomously (independently) in host cells, it must have an *ori* sequence. Without *ori*, the foreign DNA cannot maintain itself through successive cell divisions and will be diluted out.

Without *ori*, the foreign DNA cannot persist through cell divisions.

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158. (3)

List-I	List-II
Bone marrow	Production of all blood cells
Thymus	Site of T-cell maturation
Spleen	Filter of blood, stores RBCs
Lymph nodes	Traps antigens from tissue fluid

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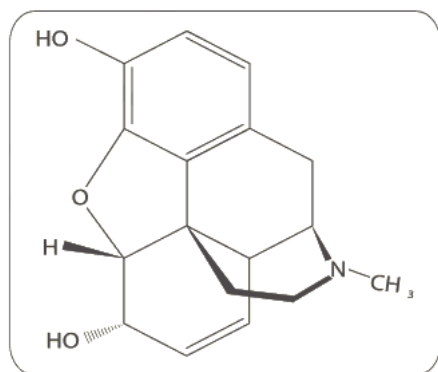
159. (1)

- Golden rice has increased Vitamin A content due to transgenes expressing carotenoid biosynthesis enzymes.
- GM crops show increased tolerance to abiotic stresses like drought and salt through transgenes for stress-response proteins.
- *Bt* crops reduce reliance on chemical pesticides through *Bt* toxin expression. However, not all GM organisms produce identical traits—transformation method, genetic background, and expression levels vary.
- GM crops with improved nutrient uptake may need less fertilizer.

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160. (3)

The chemical structure shown represents morphine.



It interacts with opioid receptors in the brain and gut, is an effective sedative and painkiller, and is extracted from the poppy plant latex. Heroin is chemically diacetylmorphine, which is a white, odourless, and bitter crystalline compound while morphine is not. Morphine is a very effective sedative and painkiller useful for patients who have undergone surgery.

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161. (2)

Scaling up human growth hormone production from laboratory to industrial level requires maintaining optimal conditions for bacterial growth and protein expression. Stirred-tank bioreactors provide precise control of temperature (37°C for *E. coli*), pH, dissolved oxygen levels, and nutrient concentration. These conditions allow high cell density cultivation and sustained hGH expression. Simple flask multiplication wouldn't achieve the same efficiency, antibiotics prevent contamination but don't improve production, and more enzymes don't increase product yield.

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162. (2)

The correct statement is that allergic antibodies are not primarily IgG type—they are IgE (immunoglobulin E) antibodies. IgE binds to mast cells and basophils, and when cross-linked by allergens, triggers histamine release causing allergy symptoms. Allergies result from overreaction to harmless substances, histamine from mast cells causes symptoms, and auto-immune diseases result from immune attack on self-antigens.

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163. (1)

List-I (PCR component)	List-II (Function)
<i>Taq</i> polymerase	Catalyzes DNA synthesis
Primers	Bind to specific target sequences
Deoxyribonucleotides	Building blocks for new DNA strands
Genomic DNA	Provides DNA template

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164. (2)

A pregnant woman with AIDS (low CD4⁺ count indicating severe immunosuppression and high viral load indicating active viral replication) faces serious risk of transmitting HIV to her unborn child. HIV can cross the placental barrier, especially during immune system compromise. This is called vertical transmission or mother-to-child transmission. Modern antiretroviral therapy dramatically reduces this risk, but without treatment, transmission probability is significant. This is a critical public health concern.

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165. (3)

Competent cells have increased cell membrane permeability, allowing DNA uptake.

Heat shock (brief exposure to high temperature like 42°C) facilitates DNA uptake in the chemical competence method by making cell membranes more permeable.

Selectable markers (usually antibiotic resistance genes) allow identification and selection of transformed cells from non-transformed cells. All three statements correctly describe transformation and gene expression processes.

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166. (3)

The transgenic mice constitute over 95% of transgenic animals. It is because mice actually have a short generation time (8-10 weeks), small size (easy to house), and well-characterized genome—these are the correct reasons for their prevalence. The reason incorrectly states "long generation time," which is false.

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167. (2)

pBR322 is specifically designed to replicate autonomously in bacteria due to its origin of replication (ori) sequence. It has multiple cloning sites (MCS), insertional inactivation of the *tet^R* gene for identifying recombinants, and both ori and *rop* sites for replication control. These features make it an excellent cloning vector precisely because it replicates independently.

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168. (1)

Antibodies consist of 2 heavy chains (2 identical heavy chains per molecule) and 2 light chains (2 identical light chains per molecule), forming a Y-shaped structure. The antigen binding sites are at the tips of the molecule, formed by portions of both heavy and light chains—not heavy chains alone. Disulphide bonds connect heavy and light chains and link heavy chains together. All antibodies have the same basic four-chain structure but different variable regions in the antigen-binding sites.

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169. (2)

Gene therapy has successfully treated certain genetic diseases but hasn't cured all genetic disorders. Retroviral vectors can integrate foreign DNA into the host genome, allowing permanent expression of the therapeutic gene in that cell and its descendants. The first clinical gene therapy actually treated ADA deficiency (adenosine deaminase deficiency), not cystic fibrosis. Gene therapy can be applied to somatic cells (affecting only the individual) and germline cells (heritable), though somatic therapy is more common.

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170. (3)

Primary immune response is slower than secondary immune response because it takes time for B cells to generate high-affinity antibodies and T cells to fully activate. Secondary response is faster because memory cells quickly recognize the pathogen and rapidly differentiate into effector cells. The primary response is also weaker, while secondary response is stronger.

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171. (4)

Agarose concentration affects pore size and separation capability (true—higher concentration, smaller pores, better separation of small fragments). Duration and voltage applied affect the speed and extent of DNA migration (true—longer duration and higher voltage separate fragments further). DNA fragment size is the basis of separation—larger fragments move slower through gel pores (true). However, the type of restriction enzyme used doesn't significantly affect electrophoresis itself—what matters is the size of fragments produced, not which enzyme created them.

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172. (3)

Vaccines contain antigens or weakened/inactivated pathogens. Vaccination provides active immunity, as the body generates its own antibodies and memory cells. Hepatitis B vaccine is produced using recombinant DNA technology (gene cloning in yeast). Vaccination generates both B lymphocytes (humoral immunity) and T lymphocytes (cell-mediated immunity), not just T cells.

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173. (2)

For developing disease-resistant rice varieties, *Agrobacterium*-mediated transformation is most suitable because it allows precise insertion of known resistance genes into the rice genome. This method is efficient for dicots and monocots (including rice), allows stable integration, and produces transgenic plants with specific, predictable traits. Classical hybridization with wild rice is less predictable and may introduce unwanted traits. Chemical mutagenesis requires extensive screening. Waiting for natural mutations is impractical.

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174. (2)

Lysozyme breaks down bacterial cell walls (specifically the peptidoglycan layer). Protease removes proteins from the macromolecule mix, not all nucleic acids—it specifically digests protein contaminants. Ribonuclease (RNase) removes RNA contamination. Purified DNA precipitates in ethanol because DNA is less soluble in alcohol than in water. These steps are essential for obtaining pure DNA from cells.

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175. (4)

Malaria prevention includes vector control (eliminating mosquito breeding sites) and mosquito nets (avoiding mosquito bites).

Typhoid prevention requires food hygiene and water purification (eliminating bacterial contamination).

Pneumonia prevention involves avoiding air-borne droplets and vaccination. However, AIDS prevention is not isolation and quarantine alone. Prevention requires comprehensive strategies including safe practices, testing, antiretroviral therapy, education about transmission routes, and support for infected individuals.

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176. (1)

Downstream processing begins after fermentation when the bioreactor culture is complete.

Separation and purification use chromatography, gel filtration, and electrophoresis.

Formulation includes preservatives (preventing contamination) and excipients (inactive ingredients improving stability and administration).

However, clinical trials are absolutely required before therapeutic products reach patients—they're essential for demonstrating safety and efficacy.

Quality control testing ensures final product safety and efficacy.

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177. (2)

Cell-mediated immunity is mediated by T-lymphocytes (particularly cytotoxic T-cells and helper T-cells), not primarily by antibodies from B cells. This response doesn't involve antibody production but rather direct cellular interactions. T-lymphocytes recognize and destroy infected cells, cancer cells, and pathogens. This response develops after first exposure but is stronger on subsequent exposures. Cell-mediated immunity is particularly important for viral infections, intracellular pathogens, and tumor immunity.

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178. (2)

Obesity and diabetes are lifestyle-related, non-communicable diseases influenced heavily by diet, exercise, physical activity levels, and environmental factors. Cancer also has lifestyle components (smoking, sun exposure, diet) though it has multiple causes. Auto-immune diseases and genetic disorders like hemophilia are not primarily lifestyle-related—they result from genetic factors and immune dysregulation. Lifestyle diseases are preventable through healthy living practices.

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179. (4)

The 'Pomato' is a somatic hybrid produced by the fusion of protoplasts from a tomato plant and a potato plant. Somatic hybridization occurs through vegetative/protoplast fusion, not sexual reproduction.

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180. (3)

The diagram shows DNA spooling, an ancient molecular biology technique. When DNA precipitates from solution (typically using alcohol precipitation), it emerges as visible, thread-like strands that can be wound onto a glass rod or stirring stick. This demonstrates the physical properties of high-molecular-weight DNA and produces a visually impressive result. DNA spooling was historically used for DNA isolation before modern extraction methods became standard.



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